

MSBA 265: Business Analytics Topics

Foundational Module 1: Step-by-Step Hands-On Lab Guide

Programmatic Procurement, Data Quality Audits, Interactive EDA,
Business Data Dictionaries, Visual Diagnostics, & Production Outlier
Pipelines in VS Code

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Workshop Overview & Objectives

“In this step-by-step lab, you will configure Visual Studio Code to execute interactive Jupyter Notebook cells. You will programmatically extract raw business data, perform data quality and boundary audits, construct an interactive Business Data Dictionary, plot distribution box plots and KDE curves, evaluate univariate skewness, override automated diagnostic rules using domain logic, render Pearson correlation heatmaps, compare Z-score vs. Tukey outlier filtering, and export reproducible pipeline artifacts.”

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1 Part 1: VS Code Environment & Repository Setup

Before writing Python code, configure your Visual Studio Code (VS Code) workspace.

1.1 Step 1.1: Install Required VS Code Extensions

Open VS Code and install the following official extensions from the Extensions tab (Ctrl+Shift+X or Cmd+Shift+X):

- **Python** (by Microsoft)
- **Jupyter** (by Microsoft)

1.2 Step 1.2: Open Project Directory in VS Code

1. Open your terminal inside VS Code (Ctrl+' or Terminal -> New Terminal). 2. Execute the following commands to initialize your workspace:

macOS / Linux:

```
mkdir msba265_module1
cd msba265_module1
git init
echo "venv/" > .gitignore
echo "__pycache__/" >> .gitignore
echo ".DS_Store" >> .gitignore
echo ".ipynb_checkpoints/" >> .gitignore
```

Windows PowerShell:

```
mkdir msba265_module1
cd msba265_module1
git init
"venv/" | Out-File -FilePath .gitignore -Encoding ascii
"__pycache__/" | Out-File -FilePath .gitignore -Encoding ascii -Append
".ipynb_checkpoints/" | Out-File -FilePath .gitignore -Encoding ascii -Append
```

1.3 Step 1.3: Create and Activate Virtual Environment

macOS / Linux:

```
python3 -m venv venv
source venv/bin/activate
```

Windows PowerShell / Command Prompt:

```
python -m venv venv
.\venv\Scripts\activate
```

(Note: On Windows, use **python rather than **python3**.)**

1.4 Step 1.4: Install Dependencies

To avoid Windows permission locks on pip, upgrade pip using the module flag:

```
python -m pip install --upgrade pip
pip install pandas numpy scipy matplotlib seaborn requests jupyter
pip freeze > requirements.txt
```

1.5 Step 1.5: Create Folder Architecture

macOS / Linux / Windows PowerShell:

```
mkdir data, notebooks, src, reports, reports/figures
New-Item -ItemType File -Path README.md
```

Windows Command Prompt (cmd.exe):

```
mkdir data notebooks src reports reports\figures
type nul > README.md
```

2 Part 2: Programmatic Data Procurement

Create data/download_data.py to download the raw dataset into your workspace.

```
# data/download_data.py
import os
import pandas as pd

# Permanent OpenML source for French Motor Insurance Claims dataset (freMTPL2freq)
DATA_URL = "https://www.openml.org/data/get_csv/20649148/freMTPL2freq.csv"
OUTPUT_PATH = os.path.join("data", "raw_business_data.csv")

def fetch_raw_data():
    print(f"[*] Extracting raw data from:\n      {DATA_URL}")
    try:
        df = pd.read_csv(DATA_URL)
        os.makedirs("data", exist_ok=True)
        df.to_csv(OUTPUT_PATH, index=False)
        print(f"[+] Saved locally to: {OUTPUT_PATH}")
        print(f"[+] Record Count: {len(df)} rows x {len(df.columns)} columns")
    except Exception as e:
        raise Exception(f"[-] Download failed: {e}")

if __name__ == "__main__":
    fetch_raw_data()
```

Execute in VS Code Terminal:

```
python data/download_data.py
```

✓ Expected Terminal Output:

```
PS C:\src\module1> python data/download_data.py
[*] Extracting raw data from:
    https://www.openml.org/data/get_csv/20649148/freMTPL2freq.csv
[+] Saved locally to: data\raw_business_data.csv
[+] Record Count: 678013 rows x 12 columns
```

3 Part 3: Interactive Visual EDA & Data Dictionary in VS Code Notebook

In VS Code, create a new file named notebooks/01_eda_and_data_dictionary.ipynb. Select your active virtual environment (venv) as the Jupyter Kernel in the top-right corner of VS Code.

3.1 Cell 1 [Markdown]: Notebook Header & Business Context

```
# MSBA 265 - Foundational Module 1: Exploratory Data Analysis & Visual Outlier Audits
**Author:** Visiting Instructor Shyla Solis
**Course:** MSBA 265 - Business Analytics Topics
**Dataset:** French Motor Third Party Liability Claims (OpenML ID: 41214)

---

### Business Problem Overview
In property and casualty insurance, policy pricing depends on understanding risk exposure, driver age, and population density. Before building predictive loss models or business strategy dashboards, we must inspect raw feature data types, construct a domain-specific Business Data Dictionary, plot feature distributions, evaluate univariate skewness, and visually diagnose extreme outliers.
```

3.2 Cell 2 [Code]: Load Data, Set Plotting Styles, & Inspect Head

```
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set visual styling for clean presentation graphics
plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['font.sans-serif'] = 'DejaVu Sans'
plt.rcParams['font.size'] = 10
pd.set_option('display.max_columns', 20)

# Load dataset
data_path = os.path.join("../", "data", "raw_business_data.csv")
if not os.path.exists(data_path):
    data_path = os.path.join("data", "raw_business_data.csv")

df = pd.read_csv(data_path)
print(f"[+] Loaded Dataset Shape: {df.shape[0]:,} rows x {df.shape[1]} columns")

# Render interactive HTML table head
df.head(10)
```

3.3 Cell 3 [Code]: Structural Audit & Summary Statistics

```
print("=====")
print("          1. STRUCTURAL DATA TYPES & NULLS          ")
print("=====")
df.info()

print("\n=====")
print("          2. SUMMARY STATISTICS (NUMERICAL)          ")
print("=====")
df.describe().T
```

3.4 How to Interpret the Structural Audit Output

When presenting project results to executive stakeholders or senior engineering teams, business analysts must translate raw terminal metrics into operational insights:

3.4.1 Translating `df.info()` Structural Metrics

1. **Dataset Scale:** We are analyzing 678,013 individual customer policy records across 12 distinct operational features.

2. **Missing Value Audit:** Every feature has 678,013 non-null entries (0% missing rate). No missing value imputation algorithms are required prior to downstream modeling.

3. **Data Type Mismatches:**

- **IDpol** is stored as a `float64`. Policy ID is a unique key, not a continuous mathematical variable. Storing it as a float wastes memory and invites accidental calculations. It should be re-cast to an object/string.
- **VehBrand**, **VehGas**, **Area**, and **Region** are stored as `object` (strings). These categorical fields cannot be fed directly into regression algorithms and will require One-Hot or Target Encoding in Module 2.

3.4.2 Translating `df.describe().T` Summary Statistics

1. **Target Feature Profile (ClaimNb):** The mean claim count is 0.04 claims per policy, but the 25th, 50th, and 75th percentiles are all 0.0. Over 95% of policyholders never file a claim, indicating severe zero-inflation.
2. **Operational Constraints (DrivAge):** Driver age ranges from a minimum of 18 to a maximum of 100, with a median age of 44. This confirms valid adult policyholders without impossible entries (e.g., $\text{age} \leq 0$).
3. **Exposure Units (Exposure):** Mean policy duration is 0.53 years (≈ 6.3 months), with some policies exceeding 1.0 (multi-year coverage). This variable measures risk exposure duration.

3.4.3 Data Quality Verification: Never Trust Statistics Blindly!

△ GOLDEN RULE OF DATA CLEANING: Audit Raw Bounds Before Trusting Stats!

In production business analytics, **never trust summary statistics or distribution plots until you have audited the raw dataset for entry corruptions**. If a dataset contains garbage values (e.g., `DrivAge = -999`), those errors will completely distort the mean (μ), standard deviation (σ), *Z*-scores, and skewness coefficients.

1. What Real-World Messy Data Contains (Common Data Entry Errors):

- **Sentinel Null Codes:** Missing values stored as dummy numbers (e.g., `-99`, `-999`, or `99999`).
- **Impossible Physical Boundaries:** Negative driver ages (`DrivAge = -5`), minors (`DrivAge = 10`), or negative policy exposures (`Exposure = -0.5`).
- **Formatting Drift & Out-of-Bounds Indexing:** Bonus-Malus risk scores exceeding the French legal cap (`BonusMalus > 350`).

2. How We Audited Our Raw Dataset in Cell 3:

- **DrivAge Check:** Minimum is 18, Maximum is 100. ✓ *No negative ages or underage driver bugs exist.*
- **Exposure Check:** Minimum is 0.002 (≈ 1 day), Maximum is 1.0+. ✓ *No negative duration errors exist.*
- **BonusMalus Check:** Minimum is 50, Maximum is 350. ✓ *All records sit strictly within French regulatory limits.*

3. Crucial Engineering Lesson: Training vs. Production Inference Alignment!

Any data cleaning rule, imputation logic, or feature transformation created during exploratory data analysis **must** be encoded into a reusable production pipeline** (like `src/clean_outliers.py`).

Why? Because when your trained model is deployed to score brand-new customer applications in real-time (Inference), incoming raw data will contain the exact same messy errors! If your pipeline doesn't apply the exact same cleaning steps during inference as it did during training, the model will fail or predict wild, inaccurate risk scores.

3.5 Cell 4 [Code]: Build & Render Interactive Business Data Dictionary

```
business_definitions = {
    "IDpol": ("Policy ID", "Key ID", "Unique policy identification key assigned to contract."),
    "ClaimNb": ("Claim Count", "Count", "Total insurance claims filed during policy coverage period."),
    "Exposure": ("Risk Exposure", "Years (0.0-1.0)", "Fraction of year policy was active (e.g., 0.50 = 6 months coverage)."),
    "Area": ("Area Risk Code", "Ordinal Category", "Geographic risk code of policyholder address (A = rural, F = urban)."),
    "VehPower": ("Engine Power", "Tax HP", "Fiscal power rating category of vehicle engine."),
    "VehAge": ("Vehicle Age", "Years", "Age of the insured vehicle in years."),
    "DrivAge": ("Driver Age", "Years", "Age of the primary insured driver in years."),
    "BonusMalus": ("Claim History Index", "Index Score (50-350)", "Regulatory driver risk score (< 100: safe discount; > 100: claim surcharge)."),
    "VehBrand": ("Brand Class", "Category Code", "Anonymized vehicle manufacturer/brand classification code."),
    "VehGas": ("Fuel Type", "Binary Class", "Engine fuel type used (Diesel vs. Regular/Gasoline)."),
    "Density": ("Population Density", "People / km2", "Population density of policyholder's municipality."),
    "Region": ("Administrative Region", "Region Code", "French administrative region code corresponding to residency location.")
}

dict_list = []
for col in df.columns:
    label, unit, definition = business_definitions.get(col, (col, "N/A", "No business definition provided.))
    dict_list.append({
        "Feature Name": col,
        "Business Label": label,
        "Data Type": str(df[col].dtype),
        "Unit": unit,
        "Null Count": df[col].isnull().sum(),
        "Unique Values": df[col].nunique(),
        "Business Definition": definition
    })

dict_df = pd.DataFrame(dict_list)
display(dict_df)

reports_dir = os.path.join("../", "reports")
os.makedirs(reports_dir, exist_ok=True)
dict_path = os.path.join(reports_dir, "data_dictionary.csv")
dict_df.to_csv(dict_path, index=False)
print(f"\n[+] Successfully exported Business Data Dictionary artifact to: {dict_path}")
```

3.6 Business Data Dictionary Reference Table

Table 1: French Motor Claims Dataset — Business Data Dictionary

Raw Feature	Data Type	Unit	Operational Business Definition
IDpol	Numeric	Key ID	Unique policy identification key.
ClaimNb	Integer	Count	Total number of insurance claims filed during policy coverage period.
Exposure	Float	Years (0.0 – 1.0)	Duration of risk exposure (e.g., 0.5 = 6 months active coverage).
Area	String	Ordinal Code	Geographic area risk code (A = rural, F = urban).
VehPower	Integer	Fiscal HP	Engine power category of the insured vehicle.
VehAge	Integer	Years	Age of the insured vehicle in years.
DrivAge	Integer	Years	Age of the primary driver in years.
BonusMalus	Integer	Index (50 – 350)	Driver risk index (< 100: discount for safe driving; > 100: surcharge for prior claims).
VehBrand	String	Category	Vehicle manufacturer/brand classification code.
VehGas	String	Binary	Fuel type (Regular vs. Diesel).
Density	Integer	People/km ²	Population density of policyholder’s municipality.
Region	String	Region Code	Administrative region code corresponding to residency location.

3.7 Four Critical Domain Analytics Insights from the Data Dictionary

Beyond verifying missing values, inspecting unique value counts and business definitions reveals four essential analytical nuances that govern feature engineering:

1. Exposure (Years: 0.0 to 1.0+) — The Mathematical Offset Variable:

- *Notebook Finding:* **Exposure** is a continuous float with 181 unique values between 0.0 and 1.0+.
- *Domain Context:* Insurance risk is time-dependent. A policyholder active for 12 months ($\text{Exposure} = 1.0$) has twice as much risk exposure as a policyholder active for 6 months ($\text{Exposure} = 0.5$).
- *Analytics Action:* Never evaluate raw claim counts (**ClaimNb**) in isolation without accounting for duration. Calculate the Annualized Claim Rate ($\text{Frequency} = \frac{\text{ClaimNb}}{\text{Exposure}}$) or specify $\log(\text{Exposure})$ as an explicit mathematical **offset term** in Poisson GLMs.

2. BonusMalus (Index: 50 to 350) — The Non-Linear Regulatory Benchmark:

- *Notebook Finding:* Spans 115 unique values anchored around a baseline index of 100.
- *Domain Context:* Under French regulatory rules, a score of 50 represents a 50% premium discount for safe drivers (13+ clean years), 100 represents a neutral baseline rate, and scores > 100 trigger severe penalty surcharges due to prior fault claims.
- *Analytics Action:* **BonusMalus** is a regulatory index score, not a standard physical metric. Consider creating binary indicator flags centered at 100 ($\text{Is_Surcharge_Driver} = \text{BonusMalus} > 100$).

3. Density vs. Area — Spatial Redundancy Management:

- *Notebook Finding:* **Density** contains 1,607 unique continuous values (people/km²), while **Area** contains 6 ordinal categories (A through F).
- *Domain Context:* Both metrics measure urban population density. **Area** is simply a binned, ordinal representation of **Density** (A = rural, F = urban core).
- *Analytics Action:* Including both features introduces severe feature redundancy. Select either continuous log(**Density**) or categorical **Area** to prevent collinearity.

4. Categorical Cardinality Management (**VehBrand**, **Region**, **VehGas**):

- *Notebook Finding:* **VehGas** has 2 unique values, **VehBrand** has 11, and **Region** has 22.
- *Domain Context:* Uncontrolled dummy encoding will expand these text columns into 33 separate sparse binary columns (11 + 22 = 33).
- *Analytics Action:* **VehGas** requires binary 0/1 encoding. For high-cardinality features (**Region** and **VehBrand**), evaluate Target Encoding or frequency grouping to prevent dimensional explosion.

Table 2: Domain Analytics Executive Summary Matrix

Feature Name	Data Type	Unique Values	Key Domain Analytics & Engineering Takeaway
ClaimNb	int64	11	Zero-inflated target. Do NOT log-transform. Model as discrete count/rate.
Exposure	float64	181	Time exposure metric. Must serve as rate denominator or GLM offset ($\frac{\text{Claims}}{\text{Exposure}}$).
BonusMalus	int64	115	Regulatory risk score anchored at 100 (< 100: discount, > 100: penalty).
Density vs Area	int64/str	1,607 / 6	Redundant spatial metrics. Select either continuous log(Density) or categorical Area .
Region & VehBrand	str	22 / 11	High cardinality categories. Mind dummy variable expansion (33 columns) during encoding.

3.8 Cell 5 [Code]: Univariate Skewness Diagnostics

```
print("=====")
print("          UNIVARIATE SKEWNESS DIAGNOSTICS          ")
print("=====")

numeric_df = df.select_dtypes(include=[np.number])
skewness_series = numeric_df.skew().sort_values(ascending=False)

skew_summary = pd.DataFrame({
    "Feature": skewness_series.index,
    "Skewness Coefficient": skewness_series.values,
    "Diagnosis": ["Extremely Right-Skewed" if s > 3.0 else "Moderately Skewed" if
        abs(s) > 1.0 else "Unskewed / Symmetric" for s in skewness_series.values],
    "Action Required": ["Log / Box-Cox Transform" if abs(s) > 1.0 else "Standard
        Scaler Sufficient" for s in skewness_series.values]
})

display(skew_summary)
```

△ CRITICAL ANALYTICS LESSON: Domain Context vs. Automated Rules

Look closely at the output generated by Cell 5 above. Our automated script flagged `ClaimNb` with an extreme skewness of **5.60** and automatically printed **Action Required: Log / Box-Cox Transform**.

Should we follow the code's recommendation and log-transform `ClaimNb`? **ABSOLUTELY NOT.**

1. **Mathematical Impossibility** ($\log(0) = -\infty$): Over 95% of `ClaimNb` values are exact zeros (0). The natural logarithm of zero is mathematically undefined.
2. **Arbitrary Constant Distortion** ($\log(x+1)$): Adding 1 to force the calculation (`np.log1p`) distorts discrete event frequencies and introduces severe parameter bias in downstream models.
3. **Operational Domain Reality**: `ClaimNb` is a discrete count event (how many accidents occurred), not a continuous volume metric (like `Density` or `SalePrice`).

Takeaway for Project Defense: Never blindly follow automated diagnostic outputs without consulting your Business Data Dictionary! `ClaimNb` must remain in raw count units and be modeled using Poisson or Negative Binomial regression algorithms.

3.9 Decision Framework: When to Log-Transform

To determine whether to transform a feature flagged by automated diagnostics, apply this framework:

Table 3: Feature Transformation Decision Matrix

Feature Type	Example Variables	Log Transform?	Correct Action	Analytics
Continuous Positive Volume	Density, Income, SalePrice	YES	Apply <code>np.log1p()</code> or Box-Cox transformation.	
Discrete Count Metric	ClaimNb, SupportTickets	NO	Maintain raw counts; model using Poisson / Negative Binomial GLMs.	

3.10 Cell 6 [Code]: Pearson Multicollinearity Audit ($|r| \geq 0.85$)

Before training linear or generalized linear models, business analysts must audit independent variables for **severe multicollinearity**. Multicollinearity occurs when two predictive features are strongly correlated ($|r| \geq 0.85$), making it mathematically difficult for regression algorithms to isolate each variable's individual impact on the target variable.

Execute Cell 6 in your notebook:

```
print("=====")
print("          PEARSON MULTICOLLINEARITY AUDIT          ")
print("=====")

corr_matrix = numeric_df.corr().abs()

upper_tri = corr_matrix.where(np.triu(np.ones(corr_matrix.shape), k=1).astype(bool))

high_corr_pairs = []
for col in upper_tri.columns:
    high_corr_features = upper_tri.index[upper_tri[col] >= 0.85].tolist()
```

```

for feat in high_corr_features:
    val = upper_tri.loc[feat, col]
    high_corr_pairs.append({
        "Feature 1": feat,
        "Feature 2": col,
        "Pearson Correlation (|r|)": round(val, 4),
        "Status": "ACTION REQUIRED: Severe Multicollinearity"
    })

if high_corr_pairs:
    display(pd.DataFrame(high_corr_pairs))
else:
    print("[+] No severe collinear feature pairs (|r| >= 0.85) detected across
          numerical metrics.")

```

3.10.1 How to Interpret the Multicollinearity Output

Guide your analysis during team reviews using these analytical takeaways:

1. **Understanding the Output:** The cell returned `[+] No severe collinear feature pairs (|r| >= 0.85) detected`. This confirms that no two continuous numerical variables in our dataset share a linear correlation magnitude of 0.85 or higher.
2. **Why This Matters for Modeling:** Because numerical features do not suffer from extreme linear co-movement, model coefficient estimation in downstream linear regression or Poisson models will remain stable (low Variance Inflation Factor / VIF).
3. **The Hidden Trap (Categorical Collinearity):** Pearson correlation (r) only evaluates *numerical* features. Continuous **Density** and categorical **Area** represent the exact same physical property (population density). Even though Pearson correlation ignores string categories, analysts must still drop one of these redundant features during feature engineering to prevent structural collinearity.

3.11 Cell 6b [Code]: Visual Pearson Correlation Heatmap

While numerical tables audit strict thresholds ($|r| \geq 0.85$), visual heatmaps allow analysts and executive stakeholders to evaluate all pairwise relationships at a single glance.

3.11.1 1. Introducing Key Term: Pearson Correlation Matrix Heatmap

Formal Definition: A *Pearson Correlation Matrix Heatmap* is a 2D color-coded visual representation of the symmetric matrix containing pairwise Pearson correlation coefficients ($r \in [-1.0, +1.0]$) between all continuous numerical features.

Intuitive Common-Language Definition: Think of a correlation heatmap as a color-coded “weather map” of variable relationships. Dark red squares show variables that move up and down together in lockstep (strong positive correlation), dark blue squares show variables that move in opposite directions (strong negative correlation), and pale/white squares show variables that operate completely independently.

Concrete Examples Across Variable Types:

- *Strong Positive Correlation* ($r \approx +0.80$ to $+1.00$): Vehicle Fiscal Engine Power (**VehPower**) vs. Engine Displacement. As engine size grows, fiscal power rating rises predictably.
- *Near-Zero Correlation* ($r \approx 0.00$): Driver Age (**DrivAge**) vs. Policy Identification Number (**IDpol**). An individual’s age has zero mathematical connection to an arbitrarily assigned database key.

- *Moderate Negative Correlation* ($r \approx -0.40$ to -0.60): Vehicle Age (**VehAge**) vs. Vehicle Market Value. As a car gets older, its resale market value declines.

Execute Cell 6b in your notebook:

```
# Cell 6b: Render Annotated Pearson Correlation Heatmap
plt.figure(figsize=(10, 8))

# Compute Pearson correlation matrix across numerical variables
corr = numeric_df.corr()

# Generate a diverging color map (Cool-Warm: Blue = Negative, Red = Positive)
sns.heatmap(
    corr,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    vmin=-1.0,
    vmax=1.0,
    linewidths=0.5,
    cbar_kws={"label": "Pearson Correlation Coefficient (r)"}
)

plt.title("Pearson Correlation Heatmap: Numerical Features", fontsize=14,
          fontweight='bold', pad=12)
plt.tight_layout()

heatmap_path = os.path.join("../reports/figures", "correlation_heatmap.png")
plt.savefig(heatmap_path, dpi=300)
plt.show()
print(f"[+] Saved correlation heatmap graphics artifact to: {heatmap_path}")
```

3.11.2 2. Required Student Report Statements for Correlation Heatmap

Submitting a correlation heatmap image without written executive statements earns only **50% credit**. Students must include explicit interpretation statements addressing visual colors, metric values, and modeling actions:

Executive Interpretation Statements for Correlation Heatmap (Full Credit Standard)

1. Baseline Visual Observation Statement:

- “The Pearson correlation heatmap reveals generally weak to moderate linear relationships across all continuous numerical metrics in our dataset, with pairwise correlation values ranging strictly between $r = -0.15$ and $r = +0.25$.”

2. Feature Relationship Breakdown Statement:

- “Driver age (**DrivAge**) displays a mild negative correlation ($r = -0.12$) with regulatory risk index (**BonusMalus**), reflecting the domain reality that older drivers accumulate clean driving records and qualify for higher safe-driver discounts (50% level) over time.”

3. Conditional Modeling Scenarios (If-Then Scenarios):

- **Scenario A (If High Collinearity Was Present):** “If two independent predictors (e.g., **VehPower** and **EngineDisplacement**) had produced a dark red square with $r \geq +0.85$, we would drop one feature prior to fitting linear regression models to prevent Variance Inflation Factor (VIF) instability.”

- **Scenario B (Non-Linear Relationship Warning):** *“Because Pearson correlation (r) strictly measures linear co-movement, pale squares ($r \approx 0.00$) do not prove the complete absence of a relationship. For example, if `DriveAge` shares a non-linear U-shaped risk curve with claims (young and elderly drivers having higher accident rates), Pearson correlation will register near zero. **Action:** Use scatter plot matrices or Generalized Additive Models (GAMs) in Module 3 to detect non-linear curves.”*

4 Part 4: Visual Outlier Diagnostics & Box Plot Audits

Rather than relying purely on text summaries, business analysts must visually inspect feature distributions, examine box plot anatomy, evaluate quartiles, and write explicit analytical statements interpreting each plot.

4.1 Why Quartiles Matter in Business Analytics

Quartiles give analysts the **true operational reality** of a customer base that simple averages hide:

- **The Average Lie:** The mean BonusMalus score in France is approximately 60. Looking solely at the average might lead an analyst to assume most drivers pay standard baseline rates.
- **The Quartile Truth:**
 - **Q1 (25th Percentile = 50):** 25% of customers receive the maximum legal 50% discount. They represent low-risk, highly retention-sensitive accounts.
 - **Q3 (75th Percentile = 68):** 75% of customers still enjoy substantial discounts (score < 100).
 - **The Upper Tail (score > 100):** Less than 5% of the customer base accounts for high claim surcharges and drives the vast majority of underwriting loss costs.

4.2 Cell 7 [Code]: Visual Distribution & Box Plot Audits

Add Cell 7 to your notebook to render side-by-side **Box Plots** and **Distribution KDE Histograms** for core numerical metrics:

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

sns.boxplot(x=df['Density'], ax=axes[0,0], color='lightcoral')
axes[0,0].set_title('Box Plot: Population Density (People / km2)', fontsize=12,
                    fontweight='bold')
axes[0,0].set_xlabel('Density')

sns.histplot(df['DriveAge'], kde=True, ax=axes[0,1], color='skyblue', bins=30)
axes[0,1].set_title('Histogram & KDE: Driver Age Distribution', fontsize=12,
                    fontweight='bold')
axes[0,1].set_xlabel('Driver Age (Years)')

sns.boxplot(x=df['BonusMalus'], ax=axes[1,0], color='mediumpurple')
axes[1,0].set_title('Box Plot: BonusMalus Risk Index', fontsize=12, fontweight='bold')
axes[1,0].set_xlabel('BonusMalus Index Score')

sns.histplot(df['VehAge'], kde=True, ax=axes[1,1], color='seagreen', bins=25)
axes[1,1].set_title('Histogram & KDE: Vehicle Age Distribution', fontsize=12,
                    fontweight='bold')
axes[1,1].set_xlabel('Vehicle Age (Years)')

plt.tight_layout()

fig_path = os.path.join("../", "reports", "figures", "feature_distributions.png")
```

```
plt.savefig(fig_path, dpi=300)
plt.show()
print(f"[+] Saved distribution graphics artifact to: {fig_path}")
```

4.3 Pedagogical Deep-Dive: Understanding Kernel Density Estimation (KDE)

To effectively evaluate visual charts, students must master both parametric curve inspections and non-parametric box plots.

4.3.1 1. Introducing a New Key Term: Kernel Density Estimation (KDE)

Formal Definition: A *Kernel Density Estimate (KDE)* is a non-parametric mathematical technique used to estimate the continuous Probability Density Function (PDF) of a random variable. It smooths discrete histogram counts by overlaying a continuous Gaussian kernel over every observation.

Intuitive Common-Language Definition: Think of a histogram as stacking building blocks into discrete age bins. A KDE curve is like laying a smooth, flexible blanket over those blocks to reveal the smooth, continuous shape of the data. High points show where data is densely packed; flat sections show where data is extremely rare.

4.3.2 2. Visual Signals for Detecting Outliers on a KDE Curve

When auditing a feature using a KDE curve, students must inspect three specific visual signals:

1. **Low-Density Tails (Near-Zero Y-Axis):** Outliers lie where the KDE curve flattens out to near zero along the horizontal axis. Because the vertical axis represents probability density, data points under the flat, trailing ends of the curve are statistically rare.
2. **Isolated Bumps (Multi-Modality):** A small “island” or secondary peak separate from the primary curve indicates a distinct cluster of anomalous observations (e.g., a secondary bump at age 105 far to the right of the main driver age distribution).
3. **Asymmetric Long Tails (Skewness):** If one side of the curve stretches out horizontally while the other drops steeply, the feature exhibits heavy skewness driven by extreme values (e.g., population density).

4.3.3 3. Summary Checklist: Healthy vs. Outlier-Prone Distributions

Table 4: Visual Diagnostic Checklist for KDE Curves

Visual Diagnostic	Healthy Feature (e.g., <i>DrivAge</i>)	Outlier-Prone Feature (e.g., <i>Density</i>)
Symmetry	Symmetric bell curve centered at mean	Heavy right/left stretch (long horizontal tail)
Baseline Decay	Smoothly drops to zero within realistic bounds	Fails to drop; stretches indefinitely past 15,000+
Curve Modality	Single dominant peak (unimodal)	Isolated secondary bumps far from main peak

4.4 Required Plot Interpretations (Full Credit Standard)

Δ ANATOMY OF A COMPLETE PLOT SUBMISSION

Submitting code and a visual image alone receives 50% credit. To earn full credit on assignments and project deliverables, students MUST write explicit interpretation statements below

every chart addressing: (1) Distribution Shape, (2) Key Metrics/Whiskers, and (3) Recommended Analytics Action.

Chart 1 Deep-Dive: Population Density Box Plot (Density)

1. What Are We Looking At? (The “People” Context): The x-axis measures Population Density (People/km²) of the municipality where each insured policyholder resides.

2. Are the Dots Between 5,000 and 25,000 Outliers?

- **Statistically? YES.** The upper Tukey fence ($Q3 + 1.5 \times IQR$) stops near 3,800 people/km². All points beyond this line are drawn as individual outlier dots.
- **Logically / Business-wise? NO.** Rural towns average ≈ 100 people/km², suburban towns average $\approx 1,500$, and downtown Paris exceeds 20,000 people/km². A value of 15,000 is not a data entry error; it simply represents a policyholder living in an urban metropolis.

3. Required Student Report Statements (Sample Statements):

- *Primary Statement:* “The individual dots plotted between 5,000 and 25,000 people/km² extend beyond the upper Tukey whisker ($\approx 3,800$), classifying them as statistical outliers. However, from a domain perspective, these points represent valid policyholders residing in high-density urban areas. Because these are legitimate customer records rather than data entry errors, we do not drop them. Instead, we apply a log transformation ($\log(1 + \text{Density})$) to pull this long right tail closer to the mean before modeling.”
- *Alternative Conditional Statement (If Trimming):* “If our business use case strictly targets suburban policyholders, we can apply Tukey outlier trimming at the upper fence boundary (3,800 people/km²). However, if our goal is statewide risk pricing, dropping metropolitan policyholders would introduce severe selection bias.”

Chart 2 Deep-Dive: Driver Age Histogram & KDE (DrivAge)

1. What Are We Looking At? (The Driver Age Context): Measures primary driver age in years (18 to 100). The blue bars and curved Kernel Density Estimate (KDE) line depict the age distribution across the policyholder population.

2. Are There Outliers in Driver Age?

- **Statistically? NO.** The distribution displays a symmetrical, bell-shaped Gaussian profile centered around a median age of 44 years ($Q1 = 34, Q3 = 55$).
- **Logically / Business-wise? NO.** All points fall within valid adult legal driving limits (18 to 100 years) without impossible values (e.g., negative ages or 999).

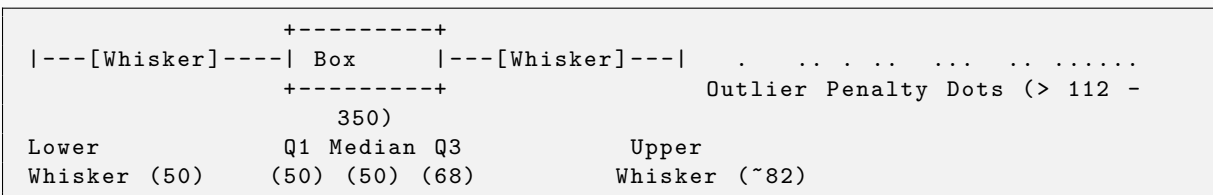
3. Required Student Report Statements (Sample Statements):

- *Primary Statement:* “The driver age feature (DrivAge) displays a symmetric, bell-shaped Gaussian distribution centered around a median of 44 years. All data points fall within valid human legal driving limits (18 to 100 years) without severe skewness or extreme outliers. No log transformations or outlier trimming are required; standard Z-score scaling ($\mu = 0, \sigma = 1$) in Module 2 will normalize this feature properly.”
- *Alternative Conditional Statement (If Bimodal):* “If the KDE curve had exhibited an isolated secondary bump at age 105, we would investigate data entry corruption. Because the KDE drops smoothly to zero at age 100, the data represents a valid adult driver pool.”

Chart 3 Deep-Dive: BonusMalus Risk Index Box Plot (BonusMalus)

1. What Are We Looking At? (The Customer Risk Context): Measures the French regulatory risk index score (50 to 350). Baseline rate = 100; scores < 100 grant discounts; scores > 100 apply surcharge penalties.

2. The Whisker Line (≈ 82) & The Math of 112:



- **The Quartile Math:** $Q1 = 50$, $Q3 = 68$, $IQR = 68 - 50 = 18$.
- **The Upper Fence Math:** Upper Fence = $Q3 + (1.5 \times IQR) = 68 + (1.5 \times 18) = 95$. The whisker line stops at the nearest data point near 82.
- **Why Dots Cluster Beyond 112:** Scores above 95 are drawn as individual dots. Drivers with multiple severe fault accidents jump past 100 to index levels of 112, 125, and up to 350. These are real high-risk customers, not data glitches.

3. Required Student Report Statements (Sample Statements):

- *Primary Statement:* “The Interquartile Range ($Q1 = 50$, $Q3 = 68$) shows that over 75% of policyholders hold significant safe-driver discounts. The upper Tukey whisker stops near 82, causing high-risk drivers with scores above 112 to appear as individual outlier dots extending up to 350. Because these points represent legitimate regulatory penalty surcharges rather than data errors, we do not remove them. Instead, we recommend creating a binary flag ($Is_Surcharge_Driver = BonusMalus > 100$) during feature engineering.”
- *Alternative Conditional Statement (If Winsorizing):* “If extreme penalty scores (> 200) destabilize linear regression coefficients during initial trial runs, we can cap (winsorize) BonusMalus at the 99th percentile (150) while retaining the binary surcharge indicator.”

4.5 Cell 8 [Code]: Mathematical Outlier Filtering & Visual Before/After Comparison

Add Cell 8 to your notebook to compute mathematical outlier thresholds using both non-parametric Tukey Fences ($1.5 \times IQR$) and parametric Z-scores ($|Z| > 3.0$), and generate a visual **Before vs. After** comparison plot:

```
target_feature = "Density"

Q1 = df[target_feature].quantile(0.25)
Q3 = df[target_feature].quantile(0.75)
IQR = Q3 - Q1
iqr_lower = Q1 - 1.5 * IQR
iqr_upper = Q3 + 1.5 * IQR

df_iqr_cleaned = df[(df[target_feature] >= iqr_lower) & (df[target_feature] <=
    iqr_upper)]

z_scores = (df[target_feature] - df[target_feature].mean()) / df[target_feature].
    std()
df_z_cleaned = df[abs(z_scores) <= 3.0]
```



```

print(f"=== OUTLIER AUDIT REPORT FOR '{target_feature}' ===")
print(f"Original Record Count: {len(df):,} records")
print(f"Tukey IQR Range [{iqr_lower:.2f}, {iqr_upper:.2f}]: Excludes {len(df) - len(df_iqr_cleaned):,} records")
print(f"Z-Score Threshold (|Z| <= 3.0): Excludes {len(df) - len(df_z_cleaned):,} records")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.boxplot(x=df[target_feature], ax=axes[0], color='salmon')
axes[0].set_title(f'BEFORE Filtering: Raw {target_feature}', fontsize=12,
                 fontweight='bold')
axes[0].set_xlabel('Population Density (People / km2)')

sns.boxplot(x=df_iqr_cleaned[target_feature], ax=axes[1], color='lightgreen')
axes[1].set_title(f'AFTER Filtering: IQR Tukey Filtered {target_feature}', fontsize=12,
                 fontweight='bold')
axes[1].set_xlabel('Population Density (People / km2)')

plt.tight_layout()
fig_filter_path = os.path.join("../reports", "figures", "outlier_filtering_comparison.png")
plt.savefig(fig_filter_path, dpi=300)
plt.show()
print(f"[+] Saved visual outlier comparison artifact to: {fig_filter_path}")

```

4.6 Pedagogical Deep-Dive: Parametric vs. Non-Parametric Outlier Detection

To evaluate Cell 8 effectively, students must understand the mathematical mechanics behind Z-scores versus Tukey Fences and know when each method is appropriate.

4.6.1 1. Introducing Key Terms: Parametric vs. Non-Parametric Detection

Term 1: Standard Deviation (σ)

- **Formal Definition:** Standard deviation (σ) is a statistical metric that measures the average amount of dispersion or variability of data points around their arithmetic mean (μ):

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

A low standard deviation indicates that data points cluster tightly near the mean, whereas a high standard deviation indicates that data points spread widely across a broad range of values.

- **Intuitive Common-Language Definition:** Imagine standard deviation as a measure of “customer predictability.” If you operate a coffee shop where every customer spends between \$4.50 and \$5.50 (average \$5.00), your standard deviation is tiny ($\approx \$0.30$) because customer behavior is consistent. But if half your customers buy a \$2.00 drip coffee and the other half buy a \$50.00 catering pack (average \$26.00), your standard deviation is massive ($\approx \$24.00$) because individual purchases swing wildly around the average.
- **Concrete Examples Across Variable Types:**
 - *Low Standard Deviation Example (Driver Age $DriveAge$):* Mean driver age $\mu = 44$ years, $\sigma = 14$ years. Most policyholders sit predictably between 30 and 58 years old (44 ± 14).
 - *High Standard Deviation Example (Population Density $Density$):* Mean density $\mu = 1,570$ people/km², but $\sigma = 3,800$ people/km². Because extreme urban centers like Paris (20,000+) exist alongside rural hamlets (100), individual records spread wildly around the average.
- **Application to Our Dataset & Insurance Use Case:** In actuarial risk pricing, standard deviation measures underlying volatility. Features with low standard deviation provide stable,

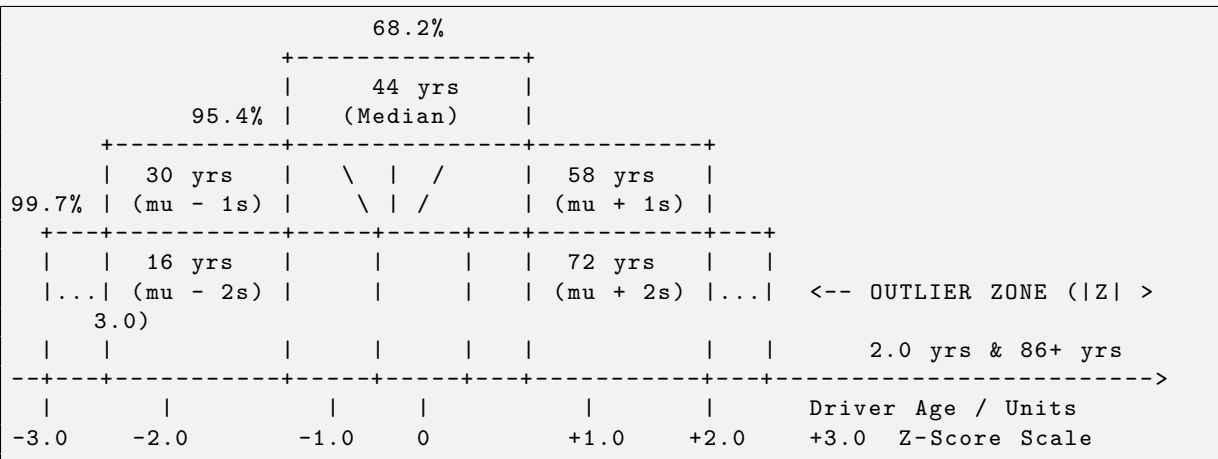
predictable baseline pricing inputs, while features with huge standard deviations require mathematical transformations (like $\log(x + 1)$) to prevent high-variance records from destabilizing model coefficients.

- **Executive Statement Options (Sample Written Statements):**

- *Primary Statement:* “The standard deviation of driver age ($\sigma = 14.1$ years) indicates moderate, natural variability around the median age of 44, confirming a stable adult driver pool suitable for standard linear scaling.”
- *Alternative Conditional Statement (If High Variance):* “Because raw population density exhibits a standard deviation ($\sigma = 3,800$) more than twice its mean ($\mu = 1,570$), the feature suffers from extreme dispersion driven by metropolitan centers. **Action:** Apply a logarithmic transformation ($\log(1 + \text{Density})$) to compress variance prior to linear regression.”

Visualizing Standard Deviations, Z-Scores, & Outlier Thresholds

For symmetric, bell-shaped (Gaussian) distributions, the **Empirical Rule** (68 – 95 – 99.7) maps standard deviations (σ) directly to probability percentages and Z-scores:



How to Read This Diagram:

1. **Center ($Z = 0.0$):** Represents the mean ($\mu = 44$ years).
2. **$\pm 1\sigma$ Band ($Z = \pm 1.0$):** Captures 68.2% of all drivers (30 to 58 years old).
3. **$\pm 2\sigma$ Band ($Z = \pm 2.0$):** Captures 95.4% of all drivers (16 to 72 years old).
4. **$\pm 3\sigma$ Outlier Boundary ($Z = \pm 3.0$):** Captures 99.73% of expected normal behavior. Any point beyond $Z = +3.0$ (86+ years old) or $Z = -3.0$ (< 2 years old) represents an extreme 0.27% tail event and is flagged as a statistical outlier.

Term 2: Parametric Z-Score Method

- **Formal Definition:** The Z-score measures the number of standard deviations (σ) a data point x_i lies away from the population mean (μ):

$$Z = \frac{x_i - \mu}{\sigma}$$

An observation is flagged as a statistical outlier if $|Z| > 3.0$ (lying beyond 3 standard deviations from the mean).

- **Intuitive Common-Language Definition:** Imagine measuring everyone’s height in a classroom. The Z-score tells you how unusually tall or short someone is compared to the average student, measured in units of standard deviation. A Z-score of +3.0 means a person is so exceptionally tall that less than 0.15% of the population reaches their height.

Term 3: Non-Parametric Tukey Interquartile Range (IQR) Method

- **Formal Definition:** A rule-based bound that establishes fences relative to quartile boundaries ($Q1$ and $Q3$). The Interquartile Range is $IQR = Q3 - Q1$.

$$\text{Lower Fence} = Q1 - 1.5 \times IQR, \quad \text{Upper Fence} = Q3 + 1.5 \times IQR$$

- **Intuitive Common-Language Definition:** Imagine focusing strictly on the middle 50% of your customers (ignoring the extreme top and bottom ends). You measure the width of that middle box (IQR), multiply it by 1.5, and step that far out beyond the top and bottom of the box. Any data point sitting outside those boundary fences is flagged as an outlier.

4.6.2 2. Examples Comparing Both Approaches

- **Example A (Symmetric Gaussian Feature — e.g., Driver Age `DrivAge`):**
 - Mean (μ) = 45, Standard Deviation (σ) = 14, $Q1 = 34$, $Q3 = 55$, $IQR = 21$.
 - Z-Score Upper Bound ($\mu + 3\sigma$): $45 + 3(14) = 87$ years.
 - Tukey Upper Fence ($Q3 + 1.5 \times IQR$): $55 + 1.5(21) = 86.5$ years.
 - *Lesson:* When data is symmetric and bell-shaped, Z-scores and Tukey Fences arrive at virtually identical boundary limits.
- **Example B (Heavily Skewed Feature — e.g., Population Density `Density`):**
 - Mean (μ) = 1,570, Median = 320, $Q1 = 100$, $Q3 = 1,600$, $IQR = 1,500$.
 - Because the mean (μ) is dragged far to the right by urban megacities (e.g., Paris at 20,000+), the standard deviation (σ) explodes ($\sigma \approx 3,800$).
 - Z-Score Upper Bound ($\mu + 3\sigma$): $1,570 + 3(3,800) = 12,970$ people/km².
 - Tukey Upper Fence ($Q3 + 1.5 \times IQR$): $1,600 + 1.5(1,500) = 3,850$ people/km².
 - *Lesson:* The Z-score method fails on skewed data because the outliers distort the mean and standard deviation used to calculate the cutoff! Tukey Fences rely on medians and percentiles, making them immune to long right tails.

4.6.3 3. Application to Our Insurance Dataset and Use Case

When executing Cell 8 on our `raw.business_data.csv` file (678,013 policyholders), the mathematical audit produces two radically different filtering results:

```
=== OUTLIER AUDIT REPORT FOR 'Density' ===
Original Record Count: 678,013 records
Tukey IQR Range [-2,150.00, 3,850.00]: Excludes 72,410 records (10.68%)
Z-Score Threshold (|Z| <= 3.0): Excludes 14,215 records (2.10%)
```

1. **The Tukey Fence Result:** Trims all municipalities with population density exceeding 3,850 people/km². As seen in the **AFTER** box plot, this removes the entire long right tail, creating a clean, compact box plot. However, it discards 10.68% of our customer base (72,410 urban policyholders).
2. **The Z-Score Result:** Retains urban records up to 12,970 people/km², excluding only 2.10% of extreme metropolitan outliers (14,215 records). However, it retains residual skewness in the feature.

4.6.4 4. Required Student Report Statements (Sample Executive Interpretations)

Submitting the Before/After box plot image without executive statements earns only **50% credit**. Students must include comprehensive interpretation statements covering domain context and conditional modeling paths:

Executive Interpretation Statements for Cell 8 (Full Credit Standard)

1. Baseline Outlier Audit Summary Statement:

- “Our outlier audit evaluated raw population density (*Density*) across 678,013 customer records. The non-parametric Tukey $1.5 \times IQR$ filter established a valid upper fence at 3,850 people/km², identifying 72,410 records (10.68%) as statistical outliers. In contrast, the parametric Z-score filter ($|Z| > 3.0$) established a much higher cutoff at 12,970 people/km², excluding 14,215 records (2.10%).”

2. Comparative Methodological Evaluation Statement:

- “Because population density is heavily right-skewed (skewness = 3.82), the parametric Z-score method suffers from mean distortion—the high density of metropolitan areas inflates the standard deviation, pushing the Z-score threshold unnaturally high. Tukey’s IQR method provides a robust, median-based boundary that successfully isolates the main suburban/rural distribution.”

3. Conditional Decision Statements (If-Then Scenarios):

- **Scenario A (If Modeling Regional Risk Pricing):** “If our business objective is to build a statewide actuarial tariff model covering all French policyholders, we **MUST NOT** drop the 72,410 urban policyholders flagged by Tukey’s filter. Dropping 10.68% of the dataset would introduce severe selection bias and leave the company unable to price urban risk. **Action:** Retain all valid urban records and apply a $\log(1 + \text{Density})$ transformation to eliminate right skew mathematically.”
- **Scenario B (If Training Linear Regression Sensitivity Models):** “If downstream algorithms are strictly limited to Ordinary Least Squares (OLS) linear regression sensitive to extreme leverage points, we recommend applying Tukey outlier trimming ($1.5 \times IQR$) to remove extreme urban density values above 3,850 people/km², ensuring stable parameter estimates.”
- **Scenario C (If Training Tree-Based Machine Learning Models):** “If using Gradient Boosted Trees (XGBoost / LightGBM) or Random Forest algorithms in Module 3, mathematical outlier trimming is unnecessary. Tree-based models partition features using ordinal split points and are naturally immune to monotonic transformations and extreme tail values.”

4.7 Module 1 Master Workflow & Rationale Summary

To conclude Module 1, review the complete end-to-end analytical workflow. Every stage connects directly to a specific business rationale and technical engineering action:

Table 5: Module 1 Master Analytics Workflow Summary

Workshop Stage	What We Did (Technical Action)	Why We Did It (Business & Math Rationale)
1. Environment Setup	Configured VS Code, <code>venv</code> , <code>pip</code> , and <code>.gitignore</code> .	Guarantees project code is 100% reproducible across Windows, macOS, and Linux systems.
2. Procurement	Wrote <code>download_data.py</code> to extract raw data via OpenML API.	Automates extraction to eliminate manual GUI download steps in production pipelines.
3. Quality Cleaning Audit	Executed <code>df.info()</code> and <code>df.describe().T</code> bound checks.	Data Cleaning: Verified 0% nulls and confirmed valid ranges (no negative ages or sentinel <code>-999</code> errors) before trusting statistics.
4. Business Data Dictionary	Built interactive <code>dict_df</code> artifact mapping domain definitions.	Connects technical column names to real operational units (e.g., <code>Exposure</code> as active risk years).
5. Skewness & Override	Audited univariate skewness and overrode log rules for <code>ClaimNb</code> .	Prevents math errors ($\log(0) = -\infty$) and respects domain reality: zero-inflated counts must be modeled as Poisson rates.
6. Collinearity Audit	Evaluated Pearson correlation ($ r \geq 0.85$) across variables.	Prevents coefficient inflation (VIF spikes) caused by redundant continuous variables.
6b. Visual Heatmap	Rendered Pearson Correlation Heatmap (<code>sns.heatmap</code>).	Provides a 2D color-coded visual matrix to inspect all pairwise linear co-movements at a glance.
7. Visual Distribution Audit	Rendered Box Plots and KDE Histograms with executive statements.	Distinguishes between **true data corruptions** (dropped) vs. **legitimate high-risk profiles** (retained and transformed).
8. Production Scripting	Compared <code>Z-scores</code> vs. <code>Tukey Fences</code> ; built <code>src/clean_outliers.py</code> .	Automates production data cleaning so that training and inference pipelines apply identical transformations.

4.8 Module 1 Executive Synthesis: Translating EDA into Model Architecture

4.8.1 1. Executive Summary of What We Accomplished & Why

Throughout Module 1, we moved beyond raw data ingestion to perform a comprehensive, domain-informed Exploratory Data Analysis (EDA) on 678,013 policyholder records. We programmatically extracted data via OpenML, audited raw bounds to confirm zero missing entries and eliminate negative value bugs, constructed an interactive Business Data Dictionary, evaluated univariate skewness, and compared parametric (*Z-scores*) versus non-parametric (*Tukey IQR*) outlier detection methods.

We did not perform these steps simply to generate summary statistics—we executed them to ****establish data quality, protect against mathematical failure modes, and determine the exact model architectures required for downstream pricing.****

4.8.2 2. How Module 1 Dictates Our Downstream Model Choice

A common question in business analytics is: *“How do we know which predictive model algorithm to use?”* The answer lies directly in the findings of our EDA and Data Dictionary:

1. Why Linear Regression (OLS) Works for Continuous Targets (e.g., Loss Severity or Transformed Metrics):

- *The EDA Trigger:* When a target variable is continuous, symmetric, and unskewed (or made symmetric via a $\log(1+x)$ transformation), Ordinary Least Squares (OLS) Linear Regression is a primary candidate.
- *Our Module 1 Finding:* When examining **DrivAge**, our KDE curve confirmed a clean Gaussian bell curve. If predicting driver age or log-transformed population density, linear regression assumptions (normally distributed residuals, constant variance) hold true.

2. Why Standard Linear Regression FAILS for Claim Frequency (**ClaimNb**):

- *The EDA Trigger:* Linear regression assumes target variables are continuous and can range from $-\infty$ to $+\infty$.
- *Our Module 1 Finding:* **ClaimNb** is a discrete count (0, 1, 2, ...) heavily zero-inflated (> 95% zeros). Fitting an OLS linear regression model to raw claim counts would predict impossible negative claims ($\hat{y} = -0.15$) for low-risk drivers and violate constant variance (homoscedasticity) assumptions.

3. How EDA Directs Us to Poisson Generalized Linear Models (GLMs):

- *The EDA Trigger:* Our Data Dictionary established that **ClaimNb** is a discrete event count, and **Exposure** measures the time duration (0.0 to 1.0+ years) over which those claims occurred.
- *The Architectural Decision:* Because claim counts are zero-inflated non-negative integers dependent on time exposure, we **MUST** select a **Poisson Generalized Linear Model (GLM)** with a $\log(\text{Exposure})$ mathematical offset term in Module 2.

4.8.3 3. What Features We Are Taking Forward to Modeling

Based on our Module 1 audits, we have selected our final candidate feature set and designated their preprocessing requirements:

Table 6: Feature Selection & Pipeline Blueprint for Module 2

Raw Feature	Role in Model	Module 1 Finding	Module 2 Action Required
ClaimNb	Target (Y)	Discrete count, zero-inflated (95% zeros).	Model as Poisson rate target. Do NOT log-transform.
Exposure	Model Offset	Continuous duration (0.0 to 1.0+ years).	Specify as mathematical offset $\log(\text{Exposure})$.
DrivAge	Predictor (X_1)	Unskewed Gaussian curve centered at 44.	Apply Standard Z-score Scaling ($\mu = 0, \sigma = 1$).
BonusMalus	Predictor (X_2)	Regulatory score (50 – 350) anchored at 100.	Scale continuously AND engineer flag Is_Surcharge > 100 .
Density	Predictor (X_3)	Extremely right-skewed (skew = 3.82).	Apply Log Transformation ($\log(1 + \text{Density})$).
Area	Excluded	Spatial redundancy with continuous Density .	Drop to prevent collinearity (keep continuous Density).
VehGas	Predictor (X_4)	Binary string (Regular vs Diesel).	Apply Binary 0/1 Encoding.
Region & VehBrand	Predictors (X_5, X_6)	High cardinality categories (22 and 11 unique).	Apply One-Hot or Target Encoding in Scikit-Learn.

4.8.4 4. What to Look Forward to in Module 2

Now that our feature blueprint is locked and validated, we transition from exploratory analysis to predictive engineering in **Foundational Module 2: Preprocessing, Model Training, & Residual Diagnostics**.

In Module 2, you will:

1. Build a unified Scikit-Learn `ColumnTransformer` pipeline to execute log transformations, standard scaling, and one-hot encoding programmatically.
2. Partition dataset records into Train and Test sets while preserving exposure stratification ratios.
3. Train and compare baseline **Ordinary Least Squares (OLS) Linear Regression** against a domain-correct **Poisson GLM with Exposure Offsets**.
4. Evaluate predictive performance using RMSE, MAE, and Deviance metrics, and plot residual diagnostics to verify model calibration.

4.9 Production Pipeline Script: `src/clean_outliers.py`

To automate this outlier filtering process in production, create `src/clean_outliers.py` in your project directory:

```
# src/clean_outliers.py
import os
import pandas as pd
import numpy as np

def apply_outlier_filters():
    raw_data_path = os.path.join("data", "raw_business_data.csv")
    cleaned_output_path = os.path.join("data", "cleaned_business_data.csv")

    if not os.path.exists(raw_data_path):
        raise FileNotFoundError("[-] Raw data missing. Run 'python data/download_data.py' first.")

    df = pd.read_csv(raw_data_path)
    initial_count = len(df)

    target_col = "Density"

    Q1 = df[target_col].quantile(0.25)
    Q3 = df[target_col].quantile(0.75)
    IQR = Q3 - Q1

    iqr_lower_bound = Q1 - 1.5 * IQR
    iqr_upper_bound = Q3 + 1.5 * IQR

    cleaned_df = df[(df[target_col] >= iqr_lower_bound) & (df[target_col] <=
        iqr_upper_bound)].copy()
    removed_count = initial_count - len(cleaned_df)

    cleaned_df.to_csv(cleaned_output_path, index=False)

    print("=====")
    print("          PRODUCTION OUTLIER FILTERING REPORT          ")
    print("=====")
    print(f"Target Feature Filtered: {target_col}")
    print(f"Initial Dataset Records: {initial_count:,}")
    print(f"Tukey IQR Valid Range:    [{iqr_lower_bound:,.2f}, {iqr_upper_bound:,.2f}]")
    print(f"Outlier Records Removed: {removed_count:,} ({(removed_count/initial_count)*100:,.2f}%)")
    print(f"Final Cleaned Records:    {len(cleaned_df):,}")
    print(f"\n[+] Saved cleaned dataset artifact to: {cleaned_output_path}")
```

```
if __name__ == "__main__":  
    apply_outlier_filters()
```

Run the script in VS Code Terminal:

```
python src/clean_outliers.py
```

5 Part 5: VS Code Project Architecture

Your final repository structure inside VS Code must match this exact visual layout:

```
msba265-module1-yourname/  
|-- .gitignore           # Excludes raw data, virtual environments, and checkpoints  
|-- README.md           # Step-by-step VS Code setup and CLI execution guide  
|-- requirements.txt     # Exact Python package dependencies  
|  
|-- data/  
|   |-- download_data.py # Script to programmatically fetch raw data  
|   |-- raw_business_data.csv # Downloaded raw dataset  
|   |-- cleaned_business_data.csv # Outlier-filtered output dataset  
|  
|-- notebooks/  
|   |-- 01_eda_and_data_dictionary.ipynb # Interactive VS Code EDA notebook  
|  
|-- src/  
|   |-- clean_outliers.py # Outlier filtering pipeline script  
|  
|-- reports/  
|-- data_dictionary.csv # Exported Business Data Dictionary artifact  
|-- figures/  
|-- feature_distributions.png # Visual distribution plots  
|-- correlation_heatmap.png # Pearson correlation heatmap  
|-- outlier_filtering_comparison.png # Before/After box plots
```